Examiner only

8. Many factors affect the rate of a chemical reaction.

(a) Explain why a change in concentration affects the rate of a reaction. [2]

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(b) Under certain conditions nitrogen monoxide reacts with carbon monoxide and oxygen according to the equation below.

$$NO(g) + CO(g) + O_2(g) \longrightarrow NO_2(g) + CO_2(g)$$

A study of the rate of this reaction, using varying initial concentrations of nitrogen monoxide and oxygen, gave the following data.

Experiment number	Concentration NO / mol dm^{-3}	Concentration O_2 / mol dm^{-3}	Initial rate of reaction / mol dm^{-3}s^{-1} × 10^{-4}
1	1.0×10^{-4}	1.0×10^{-4}	4.40
2	2.0×10^{-4}	1.0×10^{-4}	17.6
3	3.0×10^{-4}	1.0×10^{-4}	39.6
4	2.0×10^{-4}	2.0×10^{-4}	17.6

(i) Use the data to determine how the concentration of NO affects the rate of the reaction. Explain your answer by referring to the experiment numbers. [2]

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(ii) Use the data to determine how the concentration of O_2 affects the rate of the reaction. Explain your answer by referring to the experiment numbers. [2]

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(iii) Suggest a method that could be used to follow changes over time as the reaction of nitrogen monoxide, carbon monoxide and oxygen proceeds. Explain your answer. [2]

$$NO(g) \quad + \quad CO(g) \quad + \quad O_2(g) \longrightarrow NO_2(g) \quad + \quad CO_2(g)$$

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(iv) Describe the environmental implications on the atmosphere if the reaction above occurs on a large scale. [2]

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(v) The reaction shown is one of many that are catalysed in a catalytic converter. State where in a car a catalytic converter is used and name a suitable catalyst. [2]

Catalytic converter used in ....

Catalyst used ....

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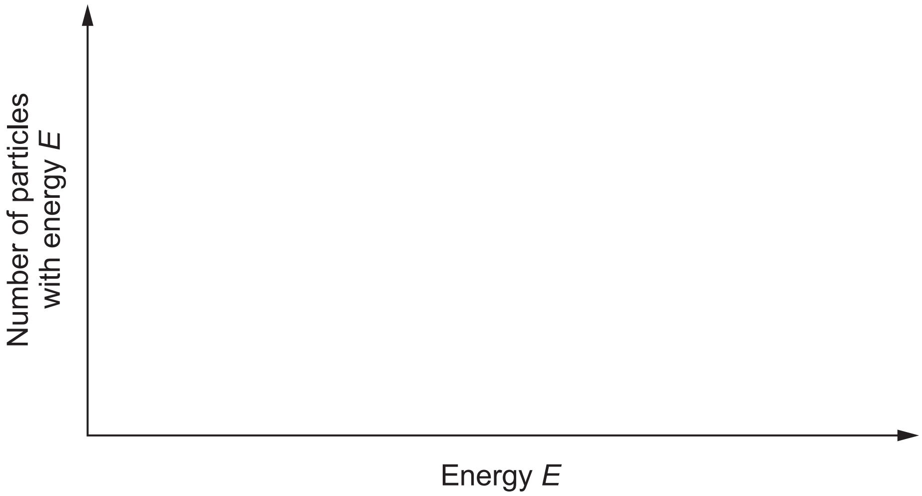
(c) On the axes below draw a Boltzmann distribution curve of the energies of molecules at a certain temperature.

Use this curve to explain how a catalyst increases the rate of a reaction. [3]



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